

FUNDAMENTALS OF DATA MINING

Lecture

Classification Techniques – Decision Tree

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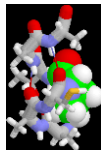
This Lecture...

- Classification Techniques
- Example of a Decision Tree
- Decision Tree Induction Algos
- Hunt's Algorithm
- Computing GINI Index

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Examples of Classification Task

- Predicting tumor cells as benign or malignant
- Classifying credit card transactions as legitimate or fraudulent
- Categorizing news stories as finance, weather, entertainment, sports, etc
- Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil



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Classification Techniques

- **Decision Tree based Methods**
- Neural Networks
- Naïve Bayes and Bayesian Belief Networks
- Support Vector Machines
- Rule-based Methods
- Memory based reasoning

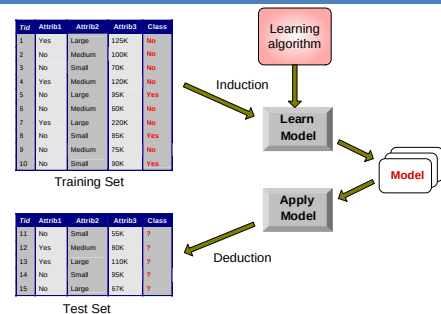
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Classification: Definition

- Goal: previously unseen records should be assigned a class as accurately as possible.
 - ▣ Given a collection of records (*training set*)
 - ▣ Each record contains a set of *attributes*, one of the attributes is the *class*.
 - ▣ Find a *model* for class attribute as a function of the values of other attributes.
 - ▣ A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

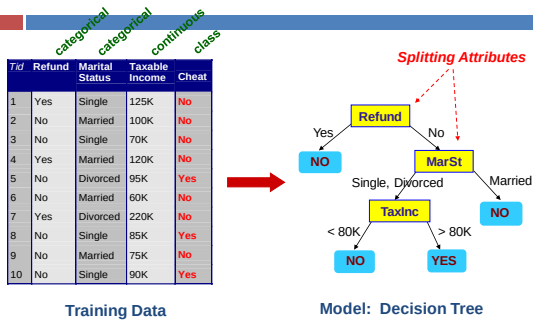
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Illustrating Classification Task



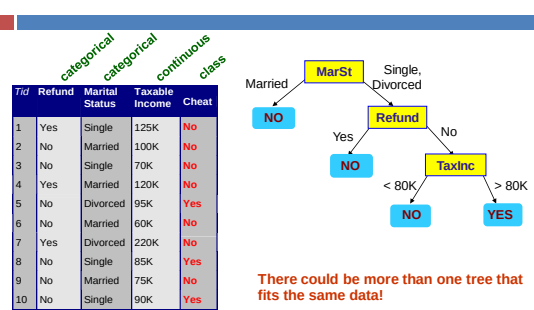
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Example of a Decision Tree



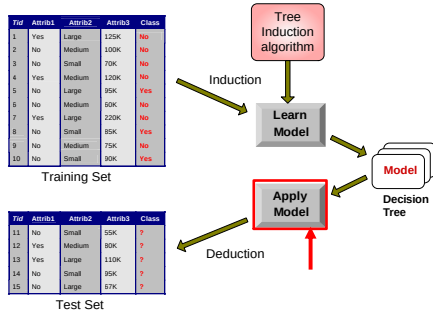
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Another Example of Decision Tree



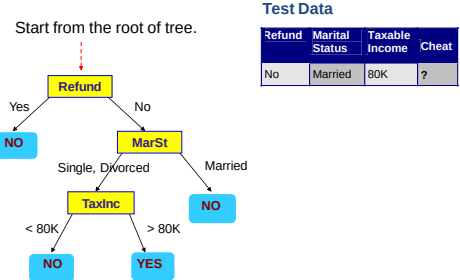
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Decision Tree Classification Task



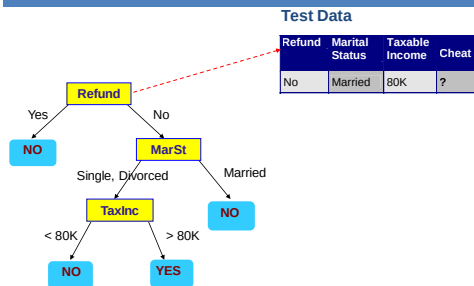
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Apply Model to Test Data



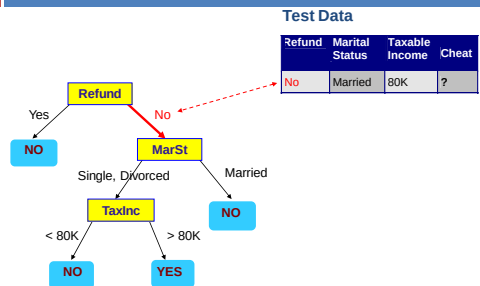
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Apply Model to Test Data



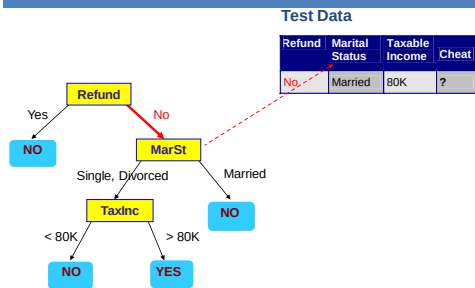
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Apply Model to Test Data



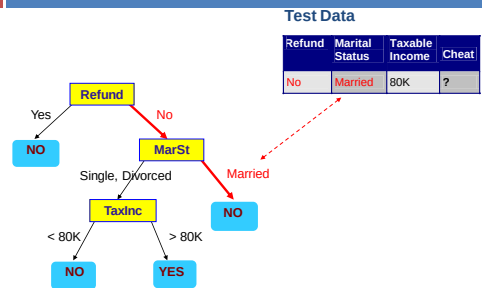
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Apply Model to Test Data



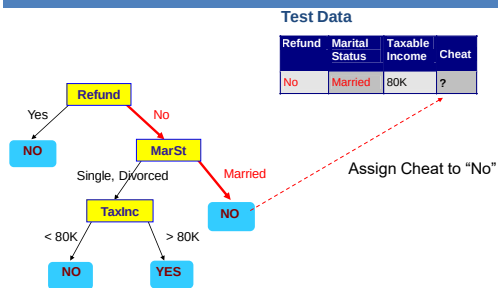
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Apply Model to Test Data



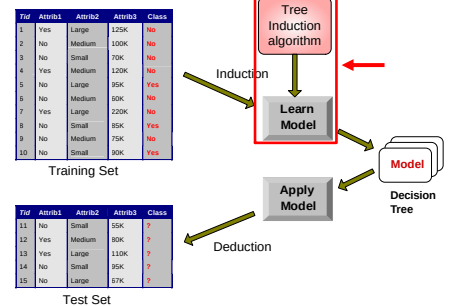
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Apply Model to Test Data



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Decision Tree Classification Task



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Decision Tree Induction

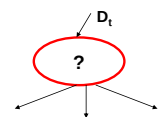
- Many Algorithms:
 - Hunt's Algorithm (one of the earliest)
 - CART
 - ID3, C4.5
 - SLIQ, SPRINT

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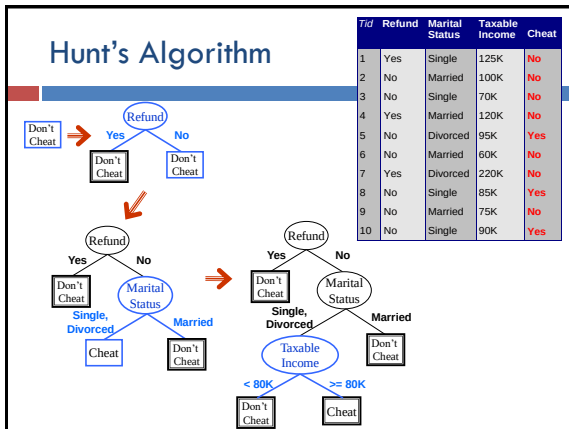
General Structure of Hunt's Algorithm

- Let D_t be the set of training records that reach a node t
- General Procedure:
 - If D_t contains records that belong to the same class y_i , then t is a leaf node labeled as y_i .
 - If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets. Recursively apply the procedure to each subset.

t/D	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	Yes
5	No	Divorced	95K	No
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



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Tree Induction

- Greedy strategy.
 - Split the records based on an attribute test that optimizes certain criterion.
- Issues
 - Determine how to split the records
 - How to specify the attribute test condition?
 - How to determine the best split?
 - Determine when to stop splitting

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Tree Induction

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How to Specify Test Condition?

- Depends on attribute types
 - Nominal
 - Ordinal
 - Continuous
- Depends on number of ways to split
 - 2-way split
 - Multi-way split

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Splitting Based on Nominal Attributes

- **Multi-way split:** Use as many partitions as distinct values.
- **Binary split:** Divides values into two subsets. Need to find optimal partitioning.

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Splitting Based on Ordinal Attributes

- **Multi-way split:** Use as many partitions as distinct values.
- **Binary split:** Divides values into two subsets. Need to find optimal partitioning.
- What about this split?

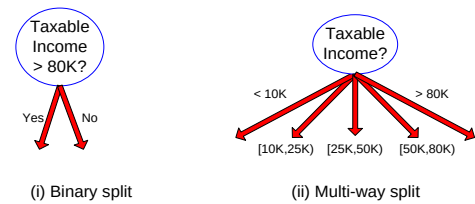
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Splitting Based on Continuous Attributes

- Different ways of handling
 - **Discretization** to form an ordinal categorical attribute
 - Static – discretize once at the beginning
 - Dynamic – ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.
 - **Binary Decision**: $(A < v)$ or $(A \geq v)$
 - consider all possible splits and finds the best cut
 - can be more compute intensive

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Splitting Based on Continuous Attributes



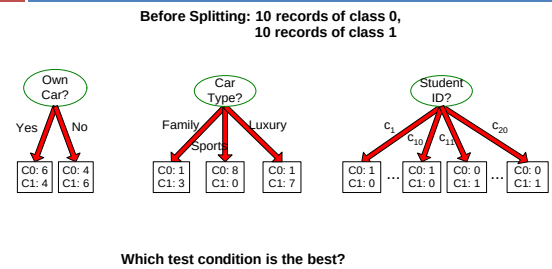
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How to determine the Best Split



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How to determine the Best Split

- Greedy approach:
 - Nodes with **homogeneous** class distribution are preferred
- Need a measure of node impurity:



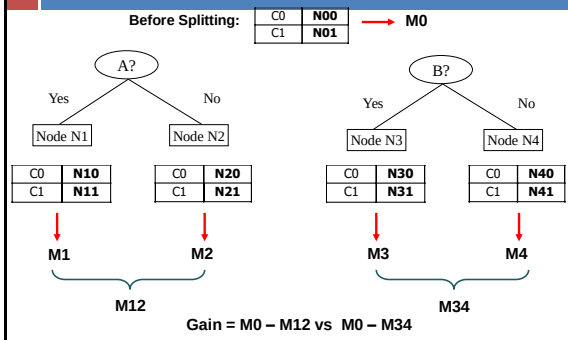
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Measures of Node Impurity

- Gini Index
- Entropy
- Misclassification error
- Oracle support both Gini and Entropy, default is Gini

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How to Find the Best Split



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Measure of Impurity: GINI

- GINI Index for a given node t :

$$GINI(t) = 1 - \sum_j [p(j|t)]^2$$

(NOTE: $p(j|t)$ is the relative frequency of class j at node t).

- Maximum ($1 - 1/n_c$) when records are equally distributed among all classes, implying least interesting information
- Minimum (0.0) when all records belong to one class, implying most interesting information

C1	0
C2	6
GINI	0.000

C1	1
C2	5
GINI	0.278

C1	2
C2	4
GINI	0.444

C1	3
C2	3
GINI	0.500

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Examples for computing GINI

$$GINI(t) = 1 - \sum_j [p(j|t)]^2$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Gini = 1 - P(C1)^2 - P(C2)^2 = 1 - 0 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Gini = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Gini = 1 - (2/6)^2 - (4/6)^2 = 0.444$$

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Splitting Based on GINI

- Used in CART, SLIQ, SPRINT.
- When a node p is split into k partitions (children), the **quality of split** is computed as,

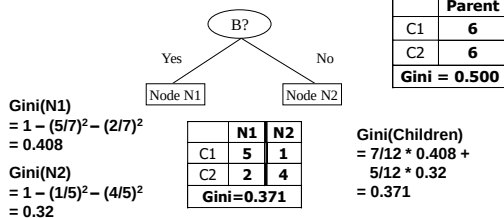
$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i)$$

where, n_i = number of records at child i ,
 n = number of records at node p .

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Binary Attributes: Computing GINI Index

- Splits into two partitions
- Effect of Weighing partitions:
 - Larger and Purer Partitions are sought for.



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Categorical Attributes: Computing Gini Index

- For each distinct value, gather counts for each class in the dataset
- Use the **count matrix** to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	2	1
C2	4	1	1
GINI	0.393		

Two-way split (find best partition of values)

	CarType {Sports, Luxury}	
	{Sports, Luxury}	{Family}
C1	3	1
C2	2	4
GINI	0.400	

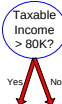
	CarType {Sports}	
	{Sports}	{Family, Luxury}
C1	2	2
C2	1	5
GINI	0.419	

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Continuous Attributes: Computing Gini Index

- Use Binary Decisions based on one value
- Several Choices for the splitting value
 - Number of possible splitting values = Number of distinct values
- Each splitting value has a count matrix associated with it
 - Class counts in each of the partitions, $A < v$ and $A \geq v$
- Simple method to choose best v
 - For each v , scan the database to gather count matrix and compute its Gini index
 - Computationally Inefficient! Repetition of work.

Id	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
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Continuous Attributes: Computing Gini Index...

- For efficient computation: for each attribute,
 - Sort the attribute on values
 - Linearly scan these values, each time updating the count matrix and computing gini index
 - Choose the split position that has the least gini index

		Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No	
		Taxable Income											
Sorted Values Split Positions		60	70	75	85	90	95	100	120	125	172	220	
		<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
Yes		0	3	0	3	0	3	1	2	2	3	0	3
No		0	7	1	6	2	5	3	4	3	4	3	5
Gini		0.420	0.400	0.375	0.343	0.417	0.400	0.300	0.343	0.375	0.400	0.420	

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Stopping Criteria for Tree Induction

- Stop expanding a node when all the records belong to the same class
- Stop expanding a node when all the records have similar attribute values

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Decision Tree Based Classification

- Advantages:
 - Inexpensive to construct
 - Extremely fast at classifying unknown records
 - Easy to interpret for small-sized trees
 - Accuracy is comparable to other classification techniques for many simple data sets

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Example: C4.5

- Simple depth-first construction.
- Uses Information Gain
- Sorts Continuous Attributes at each node.
- Needs entire data to fit in memory.
- Unsuitable for Large Datasets.
 - Needs out-of-core sorting.
- You can download the software from:
 - <http://www.cse.unsw.edu.au/~quinlan/c4.5r8.tar.gz>

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